CHANGE REPORT

✓ Fixed

Fee discount applied twice, and the exam-center form showed every institution

PROJECT School Board Panel (demo) BRANCH `fix/voucher-discount-jamia-scope` COMMIT `4f9c2e1` DATE 2026-10-28

PREPARED BY Claude (AI coding agent) REQUESTED BY Sana Iqbal, QA lead

01 Summary

Two defects turned up while the fee and exam-center screens were tested. Both are fixed on this branch.

Voucher generation took the sibling discount off twice, so 312 vouchers in the staging batch were PKR 450 too low. The new exam-center form listed every institution to nazims and saved a center for any institution id it was sent. The voucher loop now uses the amount the query has already discounted, and the form and its save action both go through the role-aware `ModuleHelper::jamiaList()`.

New feature tests cover both rules, and a person repeated the checks by hand while logged in with the least-privileged role.

AT A GLANCE

2 Problems found — 1 critical · 1 high

3 Fixes made — cover 2 of 2 problems

5 / 7 Checks passed — 1 partial · 1 not run

2 Permission requests — 1 approved · 1 safer option

1 → 0 Scan HIGH findings — before → after

02 What was wrong

Each problem as it was found, before anything was changed.

HIGH P1 **Fee discount applied twice in voucher generation**

WHERE `VoucherController::generate()` in `app/Http/Controllers/VoucherController.php` FOUND BY Test FIXED BY F1

SYMPTOM

Students with a sibling discount got vouchers below the expected amount. Example: fee 5,000 with a 10% discount should be 4,500; the voucher said 4,050.

ROOT CAUSE

The fee query already returns `net_amount` with the discount taken off. The loop that writes the `voucher_detail` rows then took the discount off a second time. Each code path assumed the other one did not apply it, and no test had a discounted student.

IMPACT

In staging batch B-17, 312 of 1,240 vouchers were PKR 450 too low: about PKR 140,400 the board would not have collected. Families would have been asked for the difference later.

EVIDENCE

> Voucher V-2026-00417: fee 5,000, discount 10%, printed 4,050.
> The query result for the same student already showed `net_amount = 4500`.

CRITICAL P2

Exam-center form shows every institution to a nazim and accepts any institution id

WHERE `ExamCenterController::create()` and `store()` FOUND BY Human FIXED BY F2 F3

SYMPTOM

A nazim, who manages one institution, opened the new exam-center form and saw all 287 enabled institutions in the dropdown. Saving with another institution selected also worked.

ROOT CAUSE

The form built its dropdown with `Jamia::where('is_enabled', 1)->get()` instead of the project helper `ModuleHelper::jamiaList()`, which applies the superadmin / nazim / staff split. The save action trusted the submitted `jamia_id` and never checked it against the user's own list.

IMPACT

Any nazim could read the names of all institutions and create exam centers for institutions they do not manage. That breaks the separation of data between institutions and can send candidates to the wrong center.

EVIDENCE

> Logged in as the test user `nazim_demo` (institution #12): the dropdown listed 287 institutions. A POST with `jamia_id=45` returned HTTP 302 and created a center for institution #45.

03 What we fixed

What changed, why it is the right fix, and the code before and after.

F1 Use the discounted amount from the query as it is

FIXES

P1 Fee discount applied twice in voucher generation

CHANGE

Removed the second discount calculation from the voucher loop. Each `voucher_detail` row now stores `net_amount` from the query. Added a feature test with one discounted and one non-discounted student.

WHY IT IS CORRECT

The discount is now taken off in exactly one place, the query, which is also what the fee register report reads, so vouchers and the register agree. The new test fails on the old code (4,050) and passes on the new code (4,500).

FILES

`app/Http/Controllers/VoucherController.php` `tests/Feature/VoucherDiscountTest.php`

− BEFORE PHP

```
    foreach ($rows as $row) {
        // net_amount comes from the query, already discounted
−       $amount = $row->net_amount
−           - ($row->net_amount * $row->discount_pct / 100);
        VoucherDetail::create([
            'voucher_id' => $voucher->id,
            'std_id'     => $row->std_id,
−           'amount'     => $amount,
        ]);
    }
```

+ AFTER PHP

```
    foreach ($rows as $row) {
        // net_amount comes from the query, already discounted
        VoucherDetail::create([
            'voucher_id' => $voucher->id,
            'std_id'     => $row->std_id,
+           'amount'     => $row->net_amount,
        ]);
    }
```

F2 Build the institution dropdown with the role-aware helper

FIXES P2 Exam-center form shows every institution to a nazim and accepts any institution id

CHANGE `create()` now calls `ModuleHelper::jamiaList()`, the helper the other forms in the project already use.

WHY IT IS CORRECT The helper implements the three-way role split (superadmin sees all, a nazim sees their own institution, staff see the institutions mapped to them), so the form follows the project rule instead of re-implementing it.

FILES `app/Http/Controllers/ExamCenterController.php`

− BEFORE PHP

```
public function create()
{
-     $jamias = Jamia::where('is_enabled', 1)->get();

    return view('exam_center.add',
compact('jamias'));
}
```

+ AFTER PHP

```
public function create()
{
+     $jamias = ModuleHelper::jamiaList();

    return view('exam_center.add',
compact('jamias'));
}
```

F3 Reject an institution id the user may not use

FIXES P2 Exam-center form shows every institution to a nazim and accepts any institution id

CHANGE `store()` validates `jamia_id` against the ids returned by `ModuleHelper::jamiaList()`. A forged id now fails validation with the normal error message.

WHY IT IS CORRECT Hiding options in a dropdown is not a security check: the server has to check the value it receives. Validation runs before any write, so nothing is saved for a forged id.

FILES `app/Http/Controllers/ExamCenterController.php` `tests/Feature/ExamCenterScopeTest.php`

− BEFORE PHP

```
$request->validate([
-     'jamia_id' => 'required|integer',
      'title'    => 'required|string|max:150',
]);
```

+ AFTER PHP

```
+ $allowed = ModuleHelper::jamiaList()->pluck('id')->all();
+
$request->validate([
+     'jamia_id' => ['required', 'integer', Rule::in($allowed)],
      'title'    => 'required|string|max:150',
]);
```

04 Verification

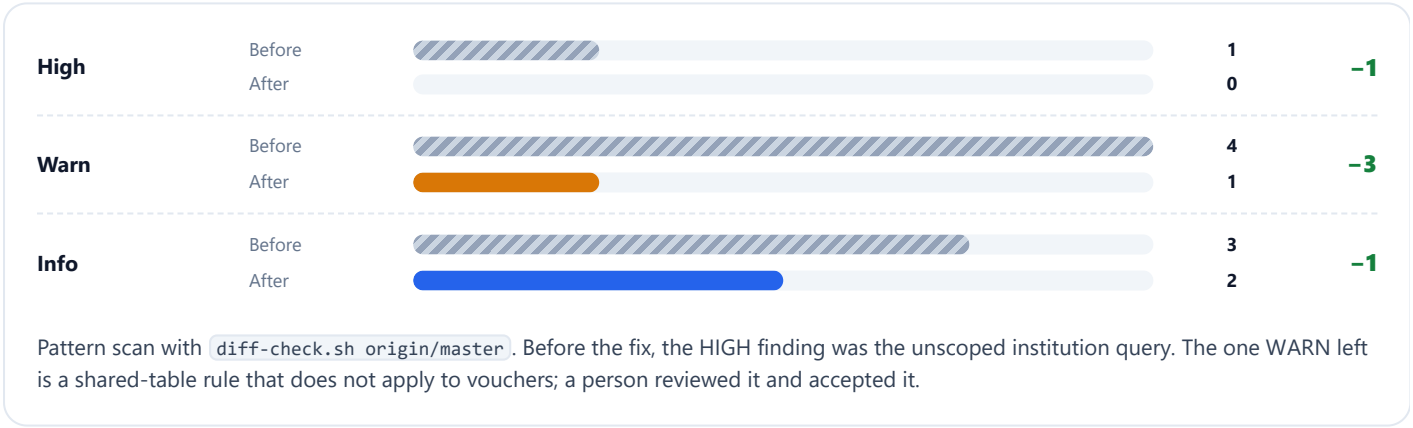
How the change was checked, by the AI agent and by people.

7 checks: 2 by the AI, 3 by people, 2 by both.

CHECK	PERSPECTIVE	RESULT	DETAIL
Feature test: a discounted student's voucher is 4,500	AI	✓ Pass	`php artisan test --filter=VoucherDiscountTest` : 2 tests, 6 assertions, on the local SQLite test database.
Batch B-17 regenerated on a local copy and compared with the fee register	AI + human	✓ Pass	312 vouchers changed, each by +450. The batch total now matches the register.
Logged in as `nazim_demo` and opened the exam-center form	Human	✓ Pass	The dropdown shows only institution #12.
Forged POST with another institution's id	Human	✓ Pass	`jamia_id=45` is rejected with a validation error; no row is created.
Superadmin and a staff user still see their full lists	AI + human	✓ Pass	Superadmin sees 287 institutions; a staff user mapped to 3 institutions sees those 3.
Other forms that build an institution dropdown	Human	◐ Partial	14 of 19 forms checked and correct. 5 still query institutions directly (see remaining risks).
Two voucher generations running at the same time	AI	○ Not run	The local setup has no queue worker. Listed as a remaining risk.

05 Scans before and after

Red-flag scan of the change, before and after the fix.



Pattern scan with `diff-check.sh origin/master`. Before the fix, the HIGH finding was the unscoped institution query. The one WARN left is a shared-table rule that does not apply to vouchers; a person reviewed it and accepted it.

06 Approvals

Risky actions the agent asked permission for, and the harm it named.

ACTION	POSSIBLE HARM	DECISION	WHEN
Run the feature tests, which reset the test database	RefreshDatabase wipes whatever database the tests point to. `.env.testing` was checked first: it points to a local SQLite file, not a shared server.	✓ Approved	2026-10-28 11:20
Regenerate voucher batch B-17 on the shared staging server. The user chose the safer option: a local copy.	Would overwrite 1,240 vouchers that other testers are using, and cannot be undone without a backup.	↪ Safer option	2026-10-28 12:05

07 Files changed

FILE	ADDED	REMOVED	SHARE
`app/Http/Controllers/VoucherController.php`	+1	−3	
`app/Http/Controllers/ExamCenterController.php`	+4	−2	
`tests/Feature/VoucherDiscountTest.php`	+64	−0	
`tests/Feature/ExamCenterScopeTest.php`	+88	−0	
Total	**+157**	**−5**	

08 Remaining risks

What is still not covered. Read these before relying on the change.

- ! Five other forms still query institutions directly. They were outside this task and are not fixed yet.
- ! Vouchers of batch B-17 printed before the fix carry the wrong amount and have to be reissued.
- ! Two voucher generations running at the same time were not tested.

09 Next steps

1. Finance approves, then the 312 affected vouchers of batch B-17 are reissued.
2. Move the five remaining forms to `ModuleHelper::jamiaList()`, one small change each.
3. Add a project rule to `.claude/safe-code-rules.tsv` so a direct institution query is flagged in every future change.
4. A second developer reads the diff before it is merged.

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Scans and tests lower the risk but do not prove that the code is safe; a person should still review every change.